

Rule-Based Frost Prediction Model with Addition of Wind Sensor and Leaf-wetness Sensor

Introduction

The rule-based model for frost prediction relies on a few fixed settings for its model's predictions. This is complimentary to the AI-based model in the analytics section of the Web app. We point out that there is a critical difference between atmospheric frost and vine injury, and our enhanced model below will clearly make that distinction as an improvement to the current model. Also, it is important to point out that our model is specialized on southwest Ohio vineyards in the April and May timeframes.

With the added leaf-wetness and wind sensors, the app can now distinguish between the two principal frost mechanisms:

- **Radiation frost:** clear, calm night; the vine canopy loses heat and becomes colder than the surrounding air.
- **Advection freeze:** a regional cold-air mass arrives with sustained wind; mixing devices are usually much less useful in this case.

So, every 15 minutes, we can now record or calculate in four different zones at the vineyard:

- Air temperature, measured by a sensor installed at the grape bud/shoot zone
- Relative humidity
- Dew point, calculated from air temperature and humidity (see Appendix A)
- Wind speed and, ideally, maximum wind gust (see Appendix B for impact)
- Leaf wetness (see Appendix B for impact)
- Soil temperature (see Appendix B for impact)
- Soil moisture (see Appendix B for impact)
- Light intensity (see Appendix B for impact)
- Temperature decline during the previous 15, 30, 60 and 120 minutes

The app should also contain several farmer-entered variables:

- Grapevine growth stage: dormant, swollen bud, bud break, 1–3-inch shoot, flowering, or fruit set

- Expected clear skies or cloudiness a couple of hours before sunrise or anticipated sky clearing during night based on weather-service cloud-cover and forecast information.
- Minimum exposure duration used in the injury model

Emerging grape tissue can be injured at temperatures near or below 32°F for as little time as 30-minute exposure, and sensitivity changes substantially with growth stage and cultivar.

The app should communicate three separate conclusions:

1. Likelihood of frost formation: Frost Watch warning/alert
2. Expected frost type: radiation, advective, or mixed
3. Likelihood of vine injury as low/medium/high based on growth stage and exposure duration

Note that frost occurring is not identical to crop damage occurring and our app should make that distinction through different levels of warning.

Web App Display and Implementation Specification

The frost warning/alert panel within the Web-app should show:

Air temperature: 34.8°F

Dew point: 34.1°F

Dew-point spread: 0.7°F

Leaf condition: Wet

Continuous wetness: 46 min

Wind speed: 1.8 mph

Surface status: Wet; surface icing possible (Type: Radiation)

Dew-Point and Leaf-Wetness Processing for Vineyard Frost Warnings/Alerts

The Web app should:

1. Calculate dew-point temperature from measured air temperature and relative humidity.
2. Retain the leaf sensor's raw measurement.
3. Derive a stable wet/dry state from the raw value.
4. Calculate wetness duration.
5. Optionally display Possible Frost/Ice, derived from wetness plus temperature conditions.

A simple wet/dry indicator is adequate for the farmer-facing screen, but wet/dry alone is not adequate for the database or rule engine. Storing the raw values for leaf-wetness would facilitate calibration, fault detection, and future model improvements.

In the calculations below, we are using fixed numbers regardless of the cultivar stage. However, we know that grapevine cold sensitivity changes sharply as buds develop. For example, Ohio State reports Pinot noir critical temperatures near 25.9°F at swollen bud, 28.0°F at bud break, 28.4°F at first leaf and 28.9°F at second leaf. Other cultivars can differ, so these numbers should not be treated as universal. Penn State likewise identifies growth stage, tissue type, surface moisture, dew point and preceding conditions as factors affecting injury. To simplify the implementation for this first phase of frost alerts, we show the fixed numbers in this document but will allow the farmer to adjust them based on cultivar and growth stage, using the provided interface when alerts are being set. These stage-specific configured values are better calibrated and provided by the farmer at this stage of development.

For this approach to benefit us long-term, every farmer-configured threshold shall be time-stamped and stored together with the vineyard zone, cultivar, growth stage, previous value, and resulting frost-event outcome so that locally calibrated settings can be evaluated retrospectively.

Derived variables

A. Temperature fall rate

Calculate:

$$CoolingRate_{60} = T_{now} - T_{60min}$$

Examples:

- -0.5°F/hour: slow cooling
- -1.0 to -2.0°F/hour: meaningful radiational cooling
- < -2.0°F/hour: rapid decline requiring escalation

Do not use a single fixed cooling-rate threshold as proof of frost. It should be combined with wind, humidity, dew point and time after sunset.

B. Dew point spread

Dew point is more useful than relative humidity alone because it estimates the temperature at which condensation begins. Define:

$$DewPointSpread = T_{air} - T_{dewpoint}$$

Interpretation:

- Greater than 8°F: relatively dry air
- 4–8°F: moderate drying potential
- 1–4°F: dew likely as temperature falls
- Less than 1°F: air is nearly saturated

A low dew point can actually produce a dangerous frost because temperature may continue falling rapidly after dew forms, while a higher dew point can temporarily slow cooling through latent-heat release. Dew point therefore affects both frost onset and likely minimum temperature.

C. Leaf-wetness duration

Track:

- Current wet/dry status
- Continuous wetness duration
- Whether wetness began without rain or irrigation

On a calm, cooling night, leaf wetness appearing as the temperature approaches dew point is strong evidence that dew is forming on vine tissue. Once air and tissue temperatures fall below freezing, that moisture may become frost. Leaf wetness is therefore a valuable confirmation sensor, but not a standalone frost predictor.

Proposed frost-warning rules

I recommend six app levels of Normal, Watch, Warning, Imminent, Event in Progress, and All Clear.

Level 1: Normal

Display green when all are true:

- Air temperature above 40°F
- No unusually rapid cooling
- Forecast minimum, when available, above 36°F
- No frost-sensitive growth stage concern

No notification is needed.

Level 2: Frost Watch

Trigger a Frost Watch when the vines are at bud swell or later and the rule set below is satisfied:

Rule 2A: Rapid dry-air cooling (provided numbers are default values adjustable by farmer in app)

- Air temperature $\leq 42^{\circ}\text{F}$

- Cooling rate $\leq -2^{\circ}\text{F}/\text{hour}$
- Wind speed < 5 mph
- Dew-point spread $> 5^{\circ}\text{F}$

Rule 2B: Forecast threshold approach

- The local forecast or projected temperature is expected to reach the configured Warning or critical threshold within six hours.

This second pattern deserves attention because dry air can permit considerable additional cooling before saturation occurs. Example notification:

Frost Watch: Vineyard temperature is falling. Current temperature 39°F , declining $1.6^{\circ}\text{F}/\text{hour}$. Continue monitoring.

Level 3: Frost Warning

Trigger a Frost Warning when frost-sensitive tissue is present and any of the following combinations persist for at least two consecutive 15-minute readings.

Rule 3A: Radiation-frost pattern (provided numbers are default values and adjustable by farmer in the app)

- Air temperature $\leq 36^{\circ}\text{F}$
- Wind speed ≤ 3 mph
- Cooling rate $\leq -1^{\circ}\text{F}/\text{hour}$
- Night-time or very low light
- Either:
 - Dew-point spread $\leq 4^{\circ}\text{F}$, or
 - Leaf wetness has changed from dry to wet

This is the classic clear/calm-night pattern. Wind machines are most useful under calm conditions when a temperature inversion exists.

Rule 3B: Rapid dry-air cooling (provided numbers are default values and adjustable by farmer in the app)

- Air temperature $\leq 37^{\circ}\text{F}$
- Wind speed ≤ 5 mph
- Cooling rate $\leq -2^{\circ}\text{F}/\text{hour}$
- Dew-point-spread $> 4^{\circ}\text{F}$ (see Appendix A)

- Projected temperature reaches 32°F within two hours

The projected temperature can initially be calculated as:

$$T_{projected,2h} = T_{now} + 2(CoolingRate_{60})$$

That is deliberately simple. Later, we can replace it with a locally calibrated regression using the vineyard's recorded nights.

Rule 3C: Wind-driven freeze approaching signaling possible Advective freeze (provided numbers are default values and adjustable by farmer in the app)

- Air temperature $\leq 36^\circ\text{F}$
- Wind speed consistently > 5 mph
- Temperature continuing to fall
- Regional weather forecast indicates freezing air

This should be labeled **Advective Freeze Warning**, not radiation frost. High winds signal an incoming cold-air mass and can make irrigation protection or high-speed fans less effective. So, in this situation we will have two distinct warning, one for radiation freeze and one for advective freeze. Example notifications are:

Radiation Freeze Warning: Bud-zone temperature is 35°F and falling. Wind is calm, and conditions favor surface frost. The configured critical temperature of 32°F may be reached within two hours. Prepare the approved frost-protection procedure.

Advective Freeze Warning: A regional freeze is approaching. Bud-zone temperature is 35°F and falling, sustained wind is 8.2 mph, and the forecast indicates temperatures below the configured critical threshold within two hours. Wind-based mixing may provide limited protection.

Level 4: Frost Imminent

Trigger Frost Imminent if any of the following occurs.

Rule 4A: Critical air-temperature approach (provided numbers are default values and adjustable by farmer in the app)

- Air temperature $\leq 33^\circ\text{F}$
- Cooling rate remains negative
- Bud break or later growth stage

Rule 4B: Frost forming on wet tissue (provided numbers are default values and adjustable by farmer in the app)

- Air temperature $\leq 33^\circ\text{F}$

- Leaf wetness is active
- Dew-point spread $\leq 1^{\circ}\text{F}$
- Wind speed ≤ 3 mph

Trigger if:

$$T_{bud_zone} \leq 32^{\circ}\text{F}$$

Rule 4C: Bud-zone temperature reaches the configured critical threshold

Trigger Frost Imminent when the latest valid bud-zone temperature reaches or falls below the farmer-configured critical temperature (provided numbers are default values and adjustable by farmer in the app)

- $T_{bud_zone} \leq 32^{\circ}\text{F}$ for the last reading

Example notification:

Frost Imminent: Measured bud-zone temperature has reached 31°F . Initiate the vineyard's frost-protection procedure immediately.

Level 5: Freeze Event in Progress

Trigger when either:

- The measured bud-zone temperature remains at or below the default value of 32°F or farmer-configured critical temperature for the farmer-configured minimum exposure duration or the default of 30 minutes; or
- The measured bud-zone temperature falls at least 2°F below the configured critical threshold in one valid reading. The 2°F margin is an adjustable default.

Display:

- Time below the configured threshold by farmer
- Time below 32°F
- Time below 30°F
- Continuous leaf wetness
- Wind classification
- Growth stage
- Vineyard zones affected

Freeze duration matters because longer exposure increases tissue injury. Cornell notes that grape shoots may survive a brief freeze but suffer more severe damage when freezing persists and ice formation disrupts tissue. Example notification:

Freeze Event in Progress: Bud-zone temperature has remained below the configured critical threshold for 30 minutes. Continue the vineyard's approved frost-protection procedure and monitoring.

Level 6: All Clear

Monitoring should continue past sunrise and then the app should report the likelihood of vine injury based on the exposure duration. Also, do not lower an alert level until:

- Temperature is at least 2°F above the applicable critical threshold;
- Temperature has been stable or rising for at least 30 minutes; and
- No forecasted decline below the critical threshold is expected during the next hour.

Issue All Clear only after those conditions are met. Also, continue logging after All Clear so the app can calculate:

- Minimum temperature
- Time below the critical threshold
- Time below 32°F
- Maximum cooling rate
- Wetness duration
- Zones affected

Example notification:

All Clear: The current frost event has ended. Bud-zone temperature is above the configured critical threshold and has remained stable or rising for 30 minutes. Continue routine monitoring.

Data-quality and sensor-failure behavior

The rules assume every sensor is operating correctly. The Web app needs to handle:

- Missing or stale measurements
- Relative humidity outside its valid range
- Frozen or stuck wind readings
- Leaf sensor permanently wet or dry
- Communication outages
- Large disagreement between zones
- Sensor-box battery or power failure

Minimum rules

- During Normal operation, mark data stale after two missed reporting intervals.
- During Frost Watch or any higher alert level, mark data stale after one missed reporting interval.
- Do not reduce the existing alert level because data have become stale
- Do not calculate dew point from invalid temperature or humidity.
- Do not downgrade an alert because a sensor stops reporting.
- If the primary temperature sensor fails during an active warning, retain the current alert and display **Sensor data unavailable**.
- Show the timestamp and age of the latest measurement.
- Store a quality flag with every derived result.

Without these rules, the system may interpret silence from a dead sensor as un-eventful night.

Two important implementation improvements

Add a sensor at vine height

The relevant air-temperature sensor should be placed near the vulnerable buds and shoots, not only inside the sensor box several feet away. Low areas should have separate sensors because cold air accumulates in frost pockets. Topography and fruiting-zone height can produce several degrees of temperature difference within the same vineyard.

Calibrate against actual events

For each frost night, retain:

- All 15-minute measurements
- Lowest canopy and air temperatures
- Duration below each threshold
- Growth stage and variety
- Observed injury percentage 24–72 hours later

After several events, replace generic thresholds with vineyard-specific ones. The strongest model will ultimately predict:

$$P(\text{damage} \mid \text{temperature}, \text{duration}, \text{growthstage}, \text{variety}, \text{wind}, \text{dew}, \text{soil})$$

That is far more useful than merely predicting whether atmospheric frost occurred.

APPENDIX A: Dew-Point and Dew-Point-Spread Calculations

Dew-Point Recommended Formula

Use the Magnus approximation:

$$\gamma = \ln \left(\frac{RH}{100} \right) + \frac{aT}{b + T}$$
$$T_d = \frac{b\gamma}{a - \gamma}$$

Where:

- T = measured air temperature in °C
- RH = relative humidity in percent
- T_d = dew-point temperature in °C
- $a = 17.625$
- $b = 243.04$ °C
- $\ln$ = natural logarithm

These coefficients are based on the Alduchov–Eskridge Magnus approximation and are suitable for ordinary near-surface agricultural temperatures. Dew point is the temperature to which air must cool to reach saturation, meaning approximately 100% relative humidity. Note that the Web app may display Fahrenheit, but dew-point calculations must be performed internally in Celsius.

Dew-Point-Spread Calculation

The app should also calculate:

$$DewPointSpread = T_{air} - T_{dewpoint}$$

This tells the app how much additional cooling is needed before air saturation and likely dew formation. Use these initial classifications for Dew-Point-Spread interpretation:

Dew-point spread	Interpretation
More than 8°F	Relatively dry air
4–8°F	Moisture increasing, but not yet near saturation
1–4°F	Dew formation may be approaching
0–1°F	Air is essentially saturated
Less than 0°F	Likely sensor error or mismatched measurements

Dew point should normally not exceed the simultaneously measured air temperature. A slightly negative spread, such as -0.2°F , may result from sensor uncertainty, but a larger negative spread should trigger a data-quality flag.

APPENDIX B: Impact of new and old sensors on the model

How the wind sensor changes the model

Wind should classify the event, not merely add points to a generic risk score.

Wind condition	Likely interpretation	Operational implication
<2 mph	Strong radiation-frost potential	Cold pockets and canopy cooling likely
2–5 mph	Radiation frost still possible	Some mixing, but risk remains
5–10 mph	Possible advective or mixed freeze	Wind machines less useful; irrigation requirements increase
>10 mph	Likely advective freeze	Regional cold-air intrusion; local mixing offers little benefit

How the leaf-wetness sensor changes the model

The leaf-wetness sensor provides four useful signals. Dew onset happens if leaves become wet while:

- No rainfall or irrigation occurred
- Wind is low
- Temperature is falling
- Air temperature is near dew point

then dew formation is likely. Frost deposition risk exists if leaf wetness is active and canopy temperature is estimated to approach 32°F , surface water may freeze.

Role of soil measurements

Soil temperature

Use soil temperature as a risk modifier, not as the main frost trigger. Increase radiation-frost risk slightly when:

- Soil temperature is unusually low for the date

- The preceding day was cold or cloudy
- Little heat was stored during daylight

Reduce risk slightly when:

- Soil is relatively warm
- The preceding day was sunny
- The surface is moist and exposed

Soil moisture

Moist soil generally stores and conducts more heat than dry soil, so dry soil can increase nighttime cooling risk. UMN identifies moist soil and bare ground as conditions that generally improve heat retention and transfer. Suggested modifier:

- Very dry soil: add one risk point
- Normal field moisture: no change
- Moist but not saturated soil: subtract one radiation-frost risk point
- Saturated soil: do not automatically reduce risk, because drainage and root-health issues become separate concerns

Role of the light sensor

The light sensor is useful for:

- Detecting local sunset
- Measuring daytime solar heating
- Identifying a cloudy versus sunny preceding day
- Estimating how much heat the vineyard soil may have stored

However, once darkness arrives, a basic light sensor cannot reliably distinguish a clear night from a cloudy night. Both look impressively dark. Cloud cover should therefore come from:

- A weather-service input, or
- A long-wave infrared sky-temperature sensor, if a fully local system is desired

A clear sky favors radiational heat loss and temperature inversion formation.